OpenClaw and local LLM subagents on 16 GB VRAM

OpenClaw is a real, explosively popular open-source AI agent platform (MIT license, ~117K GitHub stars) with native support for running multiple isolated agents against local LLMs via Ollama. It supports per-agent model selection and hot-swapping, making it viable for the one-model-at-a-time workflow on an RTX 5060 Ti 16 GB. Of the 17 models evaluated, **10 fit comfortably at 4-bit quantization, 2 fit with severe context limitations, and 5 do not fit at all**. The Qwen3 family dominates local agent tool-calling benchmarks, and most common subagent roles can be served adequately — though orchestration, complex legal/financial reasoning, and full software architecture genuinely require frontier-class models.

OpenClaw is a personal AI agent platform with multi-agent routing

OpenClaw (github.com/openclaw/openclaw) launched in late January 2026 and rapidly became one of the fastest-growing open-source projects on GitHub. Milvus Created by **Peter Steinberger** (@steipete), it is written in TypeScript/Node.js under the MIT license, with **130+ contributors** and a Discord community exceeding 12,700 members. Discord

Unlike programmatic multi-agent frameworks such as CrewAI or AutoGen (where you define agent crews in Python code), OpenClaw is a **full personal AI assistant platform** with built-in multi-agent routing. Each agent gets its own isolated workspace, state directory, session store, personality file (SOUL.md), long-term memory (MEMORY.md), and skill set. A single Gateway instance can host multiple agents, each bound to different messaging channels (WhatsApp, Telegram, Discord, Slack, Signal, iMessage, Matrix, and more). Openclaw GitHub

Key capabilities relevant to local LLM use:

- **First-class Ollama integration** with auto-discovery of tool-capable models — set OLLAMA_API_KEY="ollama-local" and it works Openclaw
- **Per-agent model selection** at runtime via the /model command (e.g., Qwen3-8B for a coding agent, Ministral 3B for a routing agent)
- **50+ bundled skills** including browser automation, shell commands, file operations, email, calendar, GitHub, and Obsidian integration agor
- **Heartbeat daemon** for autonomous scheduled tasks without user prompting Milvus
- **ClawHub** skill registry for community-built extensions GitHub
- A companion ecosystem including **openclaw-mission-control** (orchestration dashboard), **ClawRouter** (smart LLM router), and **NanoClaw** (lightweight 500-line alternative)

The project's documentation recommends **32B+ parameter models for reliable multi-step agent tasks** and notes that 8B models work for simpler automations. It requires approximately **64K token context** for full

functionality (Milvus) — a critical constraint for the 24B-class models that barely fit in 16 GB.

For comparison, the most relevant programmatic alternatives are **CrewAI** (~35K stars, Python, role-based crews), (Medium) (End Point Dev) **AutoGen/AG2** (~48K stars, Microsoft, conversational multi-agent), (GetStream +2) and **LangGraph** (LangChain ecosystem, graph-based workflows). (GetStream) All support local LLMs via OpenAI-compatible APIs. (Medium)

Which models actually fit in 16 GB at 4-bit quantization

The RTX 5060 Ti 16 GB (Blackwell, GDDR7, 448 GB/s bandwidth) (Hardware Corner) delivers roughly **51 tokens/second on 8B Q4 models** (Hardware Corner) (Hardware Corner) and supports native FP4 inference via its tensor cores. Below is the definitive VRAM assessment for every model the user listed.

Model	Total params	Active params	Q4 VRAM	Fits 16 GB?	Tool calling	Speed (est. t/s)
GPT-OSS-20B	21B MoE	3.6B	~13.7 GB	✅ Yes	✅ Excellent	~42–82
GLM-4-9B	9B	9B	~5–6 GB	✅ Yes	✅ Good	~50+
GLM-4.6V Flash 9B	9B	9B	~6–8 GB	✅ Yes	✅ Supported	~45+
Devstral Small 2 24B	24B	24B	~14–15 GB	⚠️ Tight	✅ Excellent	~20–30
Ministral 3 3B	3B	3B	~2–3 GB	✅ Yes	✅ Native	~100+
Gemma 3 27B	27B	27B	~15.6 GB	❌ No*	⚠️ Limited	N/A
Gemma 3 12B	12B	12B	~6.6 GB	✅ Yes	⚠️ Limited	~30–45
Mistral Small 3.1 24B	24B	24B	~14–15 GB	⚠️ Tight	✅ Excellent	~20–30
Phi-4 14B	14B	14B	~8–9 GB	✅ Yes	⚠️ Basic	~30–40
Phi-4-mini 3.8B	3.8B	3.8B	~2.5–3 GB	✅ Yes	⚠️ Limited	~80+
Qwen3-8B	8B	8B	~5–6 GB	✅ Yes	✅ Excellent	~51
Qwen3-4B	4B	4B	~2.5–3 GB	✅ Yes	✅ Good	~70+
DeepSeek-R1-Distill-Qwen-14B	14B	14B	~9 GB	✅ Yes	⚠️ Unstable	~30–40
Llama 3.2 3B	3B	3B	~2–3 GB	✅ Yes	⚠️ Basic	~80+
Qwen3-30B-A3B	30B MoE	3B	~17–18 GB	❌ No	✅ Excellent	N/A
Qwen3-Coder 30B-A3.3B	30B MoE	3.3B	~17–18 GB	❌ No	✅ Excellent	N/A
Nemotron 30B MoE	30B hybrid	3.5B	~17–20 GB	❌ No	✅ Excellent	N/A

Gemma 3 27B at Q4 QAT loads ~15.6 GB of weights, leaving virtually zero headroom for KV cache. Community consensus: needs 24 GB+.

Important correction: GPT-OSS-20B is actually a MoE model (21B total, 3.6B active), OpenAI not dense as categorized. Hugging Face This is why it fits so comfortably at **13.7 GB in MXFP4** — it was designed specifically for 16 GB consumer GPUs. It is the single strongest model available in this VRAM class.

The three 30B MoE models (Qwen3-30B-A3B, Qwen3-Coder 30B-A3.3B, Nemotron 3 Nano 30B) all require **17–20 GB at Q4** because every parameter must be loaded into memory regardless of activation sparsity. APXML Partial CPU offloading is possible via llama.cpp Unsloth AI but degrades speed to ~10–15 t/s — generally too slow for interactive agent use.

Comprehensive model-to-subagent matching across 40+ roles

Below is a thorough matching of community-discussed subagent roles to the models that can serve them on 16 GB. Roles are grouped by feasibility tier. For each, the **minimum quality tier** and **recommended model(s)** are stated honestly — roles that cannot be served well locally are marked clearly.

Tier 1 — Highly feasible with small models (1–4B sufficient)

These roles involve classification, routing, simple extraction, or short-form generation. They run lightning-fast on tiny models and are ideal candidates for Ministral 3 3B, Qwen3-4B, or Phi-4-mini 3.8B.

Subagent role	What it does	Recommended model(s)	Why
Router / triage agent	Classifies incoming requests and routes to the correct specialist agent	**Qwen3-4B** or **Ministral 3 3B**	Intent classification is simple; Qwen3-4B has native tool calling; ~70–100+ t/s means near-instant routing
Input validator / sanitizer	Checks inputs for format compliance, safety, completeness before passing to work agents	**Qwen3-4B** or **Phi-4-mini 3.8B**	Pattern matching and classification; doesn't need deep reasoning
Heartbeat / health monitor	Runs on a cron schedule to check system status, send alerts, perform maintenance tasks	**Ministral 3 3B**	Needs speed and reliability over intelligence; Ministral has native function calling; LM Studio OpenClaw's heartbeat daemon is designed for this atal upadhyay
Notification / alert agent	Monitors conditions and sends alerts via messaging channels	**Ministral 3 3B** or **Qwen3-4B**	Threshold checking and message formatting are simple tasks
Sentiment analysis agent	Classifies text sentiment for social media monitoring, customer feedback	**Qwen3-4B** or **Phi-4-mini 3.8B**	Classification task; small models handle this reliably
FAQ generation agent	Converts documentation into Q&A format	**Qwen3-4B**	Structured extraction + simple reformulation
Translation agent	Translates between languages	**Qwen3-8B** (for quality) or **Qwen3-4B** (common pairs)	Qwen3 models support 119 languages natively; Contabo Gradient Flow 8B gives much better quality for less-common language pairs
Calendar / scheduling agent	Manages calendar bookings, detects conflicts, coordinates meeting times	**Qwen3-4B** with tool calling	The LLM reasoning is simple; tool integration (calendar API) does the heavy lifting

Tier 2 — Feasible with mid-range models (8–14B recommended)

These roles require multi-step reasoning, summarization, or moderate tool orchestration. The **Qwen3-8B** is the workhorse here — it achieves an **F1 score of 0.919 on Docker's tool-calling benchmark** at Q4_K_M,

outperforming GPT-3.5-turbo. docker For roles needing deeper reasoning, **GPT-OSS-20B** or **Phi-4 14B** step up.

Subagent role	What it does	Recommended model(s)	Why
Research / web search agent	Gathers information using search tools, synthesizes findings into structured notes	**Qwen3-8B** or **GPT-OSS-20B**	Tool calling is critical; Qwen3-8B's excellent tool F1 and 128K context make it ideal. GPT-OSS is better for synthesis quality. Community reports good results with 8B models when paired with search tools
Content writer / blog writer	Drafts articles and long-form content from research notes	**GPT-OSS-20B** or **Qwen3-8B**	GPT-OSS gives notably better prose quality; Qwen3-8B is adequate for routine content. r/LocalLLaMA users report decent blog writing at 8B
Editor agent	Reviews content for clarity, grammar, coherence, and style	**Qwen3-8B** or **Gemma 3 12B**	Editing is well within 8B capability; Gemma 3 12B adds multimodal document review
Content planner	Plans content structure, creates outlines and editorial calendars	**Qwen3-8B**	Moderate planning; 8B handles topic analysis and outline generation well
Email drafting / summarization	Classifies incoming emails, drafts contextual responses, manages email workflows	**Qwen3-8B**	Tool calling (email API) + moderate writing quality; community reports good results at 8B
Document summarization	Condenses long documents into key points	**GPT-OSS-20B** (best quality) or **Qwen3-8B**	Summarization is one of the best local use cases; 128K context on both models handles long documents. GPT-OSS's stronger reasoning produces tighter summaries
Meeting notes / transcription agent	Processes meeting recordings, extracts summaries and action items	**Qwen3-8B**	Structured extraction + summarization; well within 8B capability. Transcription itself handled by Whisper (separate model)
Social media scouting / monitoring	Monitors social platforms, generates content, tracks trends	**Qwen3-8B**	Needs tool calling (API access) + short-form writing; 8B is plenty
Social media content agent	Generates platform-appropriate posts with hashtags and formatting	**Qwen3-4B** or **Qwen3-8B**	Short-form creative writing; even 4B handles tweets and captions

Subagent role	What it does	Recommended model(s)	Why
News aggregation agent	Collects, categorizes, and summarizes news from multiple sources	**Qwen3-8B**	Tool calling (RSS, search) + summarization; 8B handles the synthesis well
Shopping / price comparison	Compares prices across retailers, tracks deals, makes recommendations	**Qwen3-8B**	Primarily tool-driven (scraping/APIs); LLM role is comparison and recommendation
Memory / knowledge management	Curates long-term facts, decisions, and preferences for other agents	**Qwen3-8B** or **GPT-OSS-20B**	OpenClaw uses MEMORY.md files; agor this agent needs good extraction and deduplication. RAG integration benefits from Qwen3's tool calling
RAG / retrieval agent	Retrieves relevant chunks from vector stores and knowledge bases	**Qwen3-8B**	Query formulation + source citation; tool calling essential. Local embedding models (separate) handle the vector search
Data analysis agent	Cleans data, runs statistics, generates plots via code execution	**Qwen3-8B** or **Phi-4 14B**	Needs code generation (pandas/matplotlib) + reasoning about results. Phi-4's stronger STEM reasoning helps with statistical interpretation
Website scraping / understanding	Navigates websites, extracts structured information, fills forms	**Qwen3-8B** (text) or **Gemma 3 12B** (with vision)	Browser automation tools do the navigation; LLM interprets HTML and extracts data. Vision capability helps with layout understanding
Image description / understanding	Describes images, reads charts, interprets visual content	**Gemma 3 12B** or **GLM-4.6V Flash 9B**	Requires multimodal (vision+text) capability. Both fit comfortably and handle visual QA. Gemma 3 12B at QAT Q4 uses only ~6.6 GB
Customer support agent	Handles inquiries, routes to specialists, manages multi-turn support conversations	**Qwen3-8B** or **GPT-OSS-20B**	FAQ-style support works at 8B; complex issue resolution benefits from GPT-OSS's stronger reasoning
Lead scoring / prospecting	Enriches leads with company data, scores them, prioritizes for outreach	**Qwen3-8B**	Classification + tool use (CRM APIs); LLM scoring logic is straightforward

Subagent role	What it does	Recommended model(s)	Why
Sprint / project summarizer	Summarizes sprint activities and generates progress reports from PM tools	**Qwen3-8B**	Data extraction + summarization; well within 8B capability
Transaction / approval agent	Processes routine transactions, pauses for human approval on high-stakes actions	**Qwen3-4B** or **Qwen3-8B**	Classification (routine vs. high-stakes) is simple; tool execution is model-agnostic Google Developers
Home automation agent	Controls smart home devices, responds to sensor data, runs automation routines	**Qwen3-4B** with tool calling	Simple intent parsing + API calls; speed matters more than reasoning depth
Fitness / health tracking	Logs workouts, tracks nutrition, provides basic wellness summaries	**Qwen3-4B** or **Qwen3-8B**	Structured data logging + simple analysis; doesn't need deep reasoning
Recipe / cooking agent	Suggests recipes based on ingredients, adjusts portions, provides cooking guidance	**Qwen3-8B**	Moderate knowledge retrieval + structured output; 8B handles recipe generation well
Personal finance tracking	Categorizes transactions, tracks budgets, generates spending summaries	**Qwen3-8B**	Classification + basic math + structured reporting
SEO optimization agent	Analyzes content for search optimization, suggests keywords and metadata	**Qwen3-8B**	Domain-specific but not deep reasoning; keyword analysis and metadata generation
Landing page generator	Creates HTML/CSS pages from concept descriptions	**Qwen3-8B** or **Devstral Small 2 24B** (if tight fit acceptable)	Code generation task; Qwen3-8B handles basic HTML well; Devstral excels at multi-file web projects
Travel planning agent	Researches destinations, compares options, creates detailed itineraries	**GPT-OSS-20B** or **Qwen3-8B**	Multi-step tool use (search, comparison) + structured output; GPT-OSS's reasoning helps with complex multi-city itineraries

Tier 3 — Feasible but quality-limited (14B+ recommended, stretches local capability)

These roles need stronger reasoning, coding, or planning abilities. **GPT-OSS-20B** (the standout for this VRAM class), **Phi-4 14B** (for STEM reasoning), and **Devstral Small 2 24B** (for coding) are the candidates — but expect noticeable quality gaps versus frontier models.

Subagent role	What it does	Recommended model(s)	Honest assessment
Code generation agent	Writes functions, classes, and modules from specifications	**Devstral Small 2 24B** (tight fit, short context) or **GPT-OSS-20B**	Devstral hits 68% SWE-bench Verified — SOTA among open models its size. Mistral AI Limited to ~4–8K context on 16 GB. GPT-OSS is more comfortable on VRAM and handles moderate coding well. **Qwen3-8B** is a safer fallback with full context
Code review agent	Reviews code for quality, security, and maintainability	**GPT-OSS-20B** or **Devstral Small 2 24B**	Works for focused reviews of individual files. Full codebase review exceeds the context limitations of 24B models on 16 GB
QA / test writing agent	Generates unit tests and test cases from specifications	**Devstral Small 2 24B** or **Qwen3-8B**	Unit test generation for individual functions works well; test suite design for complex systems is weaker
Task planning / decomposition	Breaks complex tasks into subtasks with dependencies	**GPT-OSS-20B**	Planning quality scales strongly with model capability. GPT-OSS's reasoning (comparable to o3-mini) handles moderate task decomposition. Complex multi-dependency planning will be noticeably worse than frontier models
Fact-checking agent	Verifies claims against multiple sources, evaluates credibility	**GPT-OSS-20B** with search tools	Tool-augmented fact-checking works; the reasoning about source quality and contradiction detection benefits from GPT-OSS's stronger reasoning
Critic / review agent	Evaluates other agents' outputs against quality criteria	**GPT-OSS-20B** or **Phi-4 14B**	Quality evaluation requires good judgment; GPT-OSS handles this reasonably for straightforward criteria
Market research agent	Conducts competitive analysis, tracks trends, generates market reports	**GPT-OSS-20B** with search tools	Basic market overviews work; nuanced competitive intelligence benefits from stronger reasoning
Synthesizer / aggregator	Combines outputs from multiple agents into coherent unified responses	**GPT-OSS-20B**	Requires understanding multiple inputs and resolving contradictions; GPT-OSS handles moderate synthesis tasks

Subagent role	What it does	Recommended model(s)	Honest assessment
Curriculum / course designer	Generates lesson plans, educational materials, and instructor guides	**GPT-OSS-20B** or **Qwen3-8B**	Lesson outlines work at 8B; pedagogically sound curriculum design benefits from stronger models
Prototype generator	Rapidly builds working prototypes (Streamlit/Gradio apps) from specifications	**Devstral Small 2 24B**	Code generation specialized; handles simple app prototypes well within its context limitations
DevOps agent	Manages deployment pipelines, CI/CD, log analysis	**GPT-OSS-20B**	Shell command generation works; complex infrastructure reasoning is marginal

Tier 4 — Deep reasoning roles where DeepSeek-R1-Distill shines (but with caveats)

DeepSeek-R1-Distill-Qwen-14B (~9 GB at Q4) BytePlus delivers exceptional chain-of-thought reasoning comparable to o1-class models. However, its **tool calling is unstable** — community patches exist but produce "looped calls or empty responses." Ollama Use it exclusively for reasoning-only roles where tool calling isn't needed.

Subagent role	What it does	Recommended model	Caveats
Math / logic reasoning	Solves mathematical problems, proofs, statistical calculations	**DeepSeek-R1-Distill-Qwen-14B**	Exceptional math reasoning at this size. Set temperature 0.5–0.7 to avoid repetition. No system prompt support. Hugging Face Thinking tokens add 5–10× latency
STEM analysis	Scientific literature interpretation, technical analysis	**Phi-4 14B** or **DeepSeek-R1-Distill-Qwen-14B**	Phi-4 scores 84.8 MMLU and excels on GPQA/MATH. Better tool-calling support than DeepSeek-R1
Algorithm exploration	Generates novel algorithmic approaches, evaluates computational complexity	**DeepSeek-R1-Distill-Qwen-14B**	Community reports working on AutoGen algorithm discovery experiments. Towards Data Science Results are modest but show local feasibility Towards Data Science

Tier 5 — Not feasible with these local models (honest assessment)

These roles genuinely require frontier-level intelligence, massive context, specialized domain expertise, or real-

time capabilities that no model in the 16 GB VRAM class can provide adequately. **Do not force a match.**

Subagent role	Why it doesn't work locally	What would be needed
Orchestrator / manager agent (for complex multi-agent workflows)	Community consensus: the orchestrator needs the strongest model in the system. It must understand all sub-agent capabilities, decompose ambiguous goals, handle failures gracefully, and maintain coherent state across many turns. GPT-OSS-20B can handle simple orchestration, but complex workflows with 5+ agents and conditional branching degrade significantly	70B+ model or cloud API (Claude Sonnet, GPT-4o). Community pattern: "big model for routing, small models for execution"
Software architect agent	Translating requirements into full system architecture requires holistic reasoning across databases, APIs, scaling, security, and tradeoffs. Community reports poor results under 13B, and even 14B produces shallow architectures	70B+ or frontier API
Complex debugger agent	Multi-step error analysis across codebases with non-obvious root causes. Requires maintaining mental models of runtime state, data flow, and side effects	34B+ with long context; frontier models preferred
Legal document review	Requires deep domain expertise, understanding of jurisdiction-specific precedent, and extreme precision. Liability risk makes this inappropriate for small models regardless of capability	Frontier API + human review; not suitable for any local model at this scale
Medical / health analysis (beyond basic tracking)	Safety-critical domain where hallucinations cause real harm. Community strongly advises against relying on local models for medical interpretation	Frontier API + mandatory human oversight
Financial analysis / stock research	Deep fundamental analysis, technical analysis, and investment recommendations require strong numerical reasoning plus domain expertise. Community reports significantly better results with API models	Frontier API for serious analysis; GPT-OSS-20B handles basic financial summaries
Security analyst agent	Vulnerability analysis and security auditing require specialized knowledge that general-purpose models lack at 8–14B	Specialized security models or frontier API
SRE / incident response	Root-cause analysis across microservices with complex failure modes. Requires correlating logs, metrics, and traces across systems	Frontier API + monitoring tool integration
Book writing agent (coherent long-form)	Maintaining narrative coherence across chapters requires long-horizon planning far beyond what 8–14B models handle	70B+ with 128K+ context
Compliance / regulatory agent	Domain-specific regulatory knowledge, risk modeling, and audit-quality reasoning are beyond general local models	Frontier API + domain-specific fine-tuning

Practical integration: hot-swapping, backends, and latency

Model swapping is the dominant bottleneck on single-GPU setups

On the RTX 5060 Ti with an NVMe SSD, model loading times are approximately:

- **1–4B models**: 1–3 seconds
- **8B models**: 2–5 seconds
- **14B models**: 5–10 seconds
- **24B models** (tight fit): 10–15 seconds

A five-agent workflow using three different models adds **9–21 seconds of pure swap overhead per cycle**. Three strategies minimize this cost:

1. **Use the same model for all agents** — different system prompts customize behavior without swapping. A single Qwen3-8B instance can serve as researcher, writer, and summarizer simultaneously
2. **Keep two small models loaded simultaneously** — Ollama can hold a 4B routing model (~2.5 GB) alongside an 8B work model (~5.5 GB) within 16 GB, with room for KV cache. Set `OLLAMA_MAX_LOADED_MODELS=2`
3. **Use vLLM Sleep Mode** (v0.11.0+) — offloads model weights to CPU RAM instead of fully unloading. Wake time drops to **0.1–2.6 seconds** versus 10–45 seconds for cold starts [vLLM]

Ollama is the pragmatic backend for OpenClaw

OpenClaw has first-class Ollama integration. Configuration is minimal: export `OLLAMA_API_KEY="ollama-local"` and OpenClaw auto-discovers available models. [Openclaw] For agent workflows, set `OLLAMA_KEEP_ALIVE=-1` to prevent mid-conversation model unloads. [Restack] Ollama's default behavior keeps models loaded for 5 minutes after last use [Daniel Nashed's Blog] and supports up to **3× GPU count** models simultaneously. [Restack]

Alternative backends (llama.cpp server, vLLM, LM Studio, LocalAI) all expose OpenAI-compatible APIs [Medium] [Clawctl] and work with OpenClaw, CrewAI, AutoGen, and LangGraph. **vLLM** offers 3–35× higher throughput than Ollama under concurrency but only serves one model per instance. [Medium] **LM Studio** provides the best desktop GUI experience for model management. [Clawctl]

Tool calling is the make-or-break capability for agents

Docker's benchmark [Docker] of **21 models across 3,570 test cases** reveals a stark divide in tool-calling reliability: [Docker]

- **Qwen3-14B Q4_K_M**: F1 0.971 (nearly matches GPT-4's 0.974) docker
- **Qwen3-8B Q4_K_M**: F1 0.919 (outperforms GPT-3.5-turbo's 0.899) docker
- **Llama 3.1 8B Q4_K_M**: F1 0.793 (marginal for agent use) docker
- **Gemma 3 4B**: F1 0.733 (unreliable) docker
- **Sub-4B models**: "Significant reliability issues" — hallucinated tool names, malformed arguments, infinite loops GitHub

Qwen3-8B is the minimum viable model for reliable agent tool calling. GitHub Below this threshold, agents frequently hallucinate tool names, generate malformed arguments, or enter call loops. For roles where tool calling is non-negotiable (research, email, calendar, browser automation), Qwen3-8B or GPT-OSS-20B should be the default.

Models without native tool calling (DeepSeek-R1-Distill, base Phi-4, Llama 3.2 3B) can still participate in agent workflows via **ReAct prompting** (Thought/Action/Observation format Prompt Engineering Guide with regex parsing) or **constrained generation** (grammar-guided JSON output GitHub via llama.cpp or SGLang). These workarounds add latency and reduce reliability but make reasoning-focused models usable as non-tool-calling subagents.

The recommended one-GPU agent stack

For an RTX 5060 Ti 16 GB running OpenClaw with model hot-swapping:

- **Primary work model**: **GPT-OSS-20B** at MXFP4 (~13.7 GB) — best overall quality, excellent tool calling, designed for 16 GB GPUs, ~42 t/s. Use for research, writing, planning, coding, and most agent roles
- **Fast routing model**: **Qwen3-4B** at Q4 (~2.5 GB) — native tool calling, ~70 t/s. Use for triage, classification, heartbeat tasks, and notifications. Can coexist in VRAM alongside GPT-OSS-20B if VRAM allows after KV cache
- **Vision model** (swap in when needed): **Gemma 3 12B QAT** (~6.6 GB) or **GLM-4.6V Flash 9B** (~6–8 GB) — for image understanding, document analysis, visual QA
- **Deep reasoning model** (swap in when needed): **DeepSeek-R1-Distill-Qwen-14B** (~9 GB) — for math, logic, and STEM tasks where chain-of-thought reasoning matters more than tool use
- **Coding specialist** (swap in for intensive coding sessions): **Devstral Small 2 24B** (~14–15 GB) — tight fit, short context, but SOTA agentic coding. Only load when doing substantial code generation

This stack covers the vast majority of subagent roles with **3–4 model swaps** rather than 17 different models, keeping swap overhead manageable.

Conclusion

OpenClaw's architecture is well-suited for this single-GPU workflow. Its per-agent model selection, Ollama auto-discovery, and skill-based architecture mean you can map different subagent roles to appropriately-sized models and swap them in as needed. The **GPT-OSS-20B** is the breakout star of the 16 GB VRAM class — a MoE model with only 3.6B active parameters that fits at 13.7 GB while delivering reasoning comparable to o3-mini and excellent tool calling. Paired with **Qwen3-8B** (the tool-calling champion at its size) and specialized models for vision and reasoning, roughly **30 of 40+ community-discussed agent roles can be served adequately on this hardware**.

The honest gaps are in orchestration of complex multi-agent workflows, domain-expert roles (legal, medical, financial, security), and tasks requiring sustained coherence over very long contexts. For these, a hybrid approach — local models for the majority of agent roles, cloud API calls for the orchestrator and frontier-demanding tasks — delivers the best practical result. OpenClaw's model fallback chains and auth profile rotation make this hybrid pattern straightforward to configure.